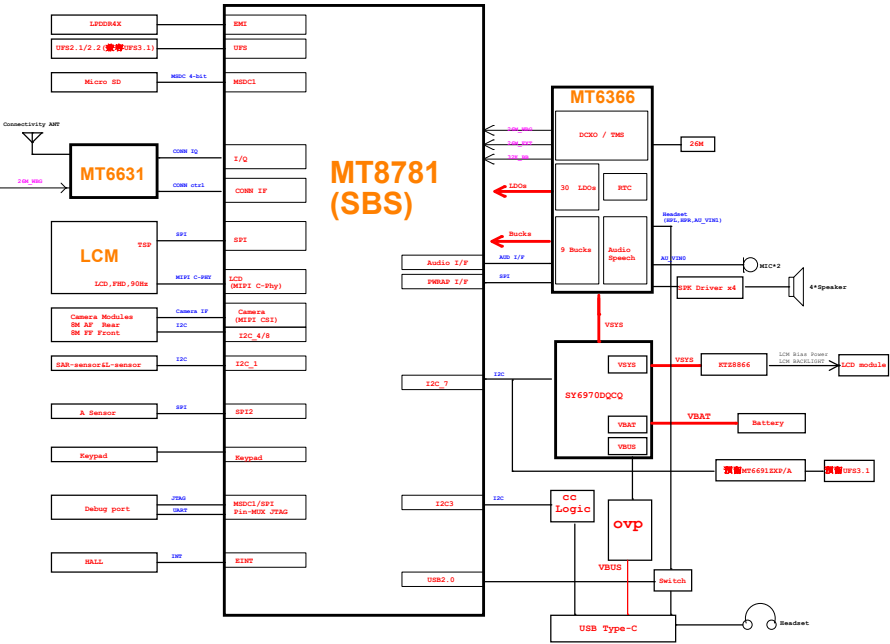
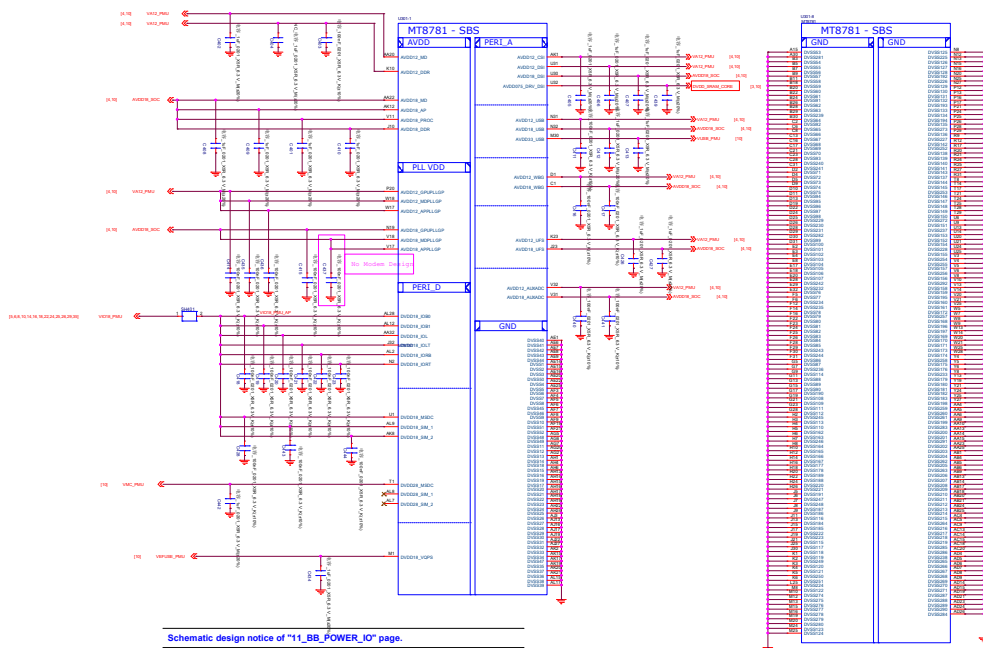




华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		01.Block Diagram	
Size	Project	L83_MB_V1	Rev
D			V1
Date: Saturday, July 02, 2022		Sheet 1 of 38	

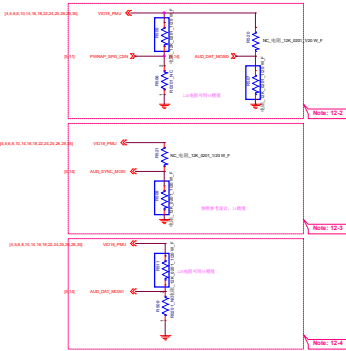
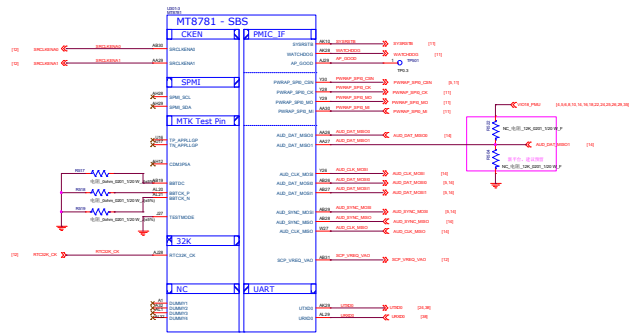
Date	Version	Page	Change notice
2021.12.15		1	修改屏连接器型号和pin数，热仿真需求增加MINTC3个。
2021.12.20		17	删除0.9V LDO
2021.12.21		25	G sensor 改为SPI通信
2021.12.24		26	OVP 改为 AH32410
2021.1.4		16	增加独立的LCM VDDI LDO 备用，NC

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title02.Change_Notice		
Size D	ProjectL83_MB_V1	Rev V1
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Schematic design notice of "11_BB_POWER_IO" page.

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		04.BB_IOVDD	
Size	Project	L83_MB_V1	Rev
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Schematic design notice of "12_BB_1" page:

Note 12-1: The de-coupling cap. for REFP have to be placed as close to BB as possible.

Note 12-2: "PWRAP_SPD_CSN" and "AUD_DAT_MOSI0" pin features in trapping pin to enable JTAG.

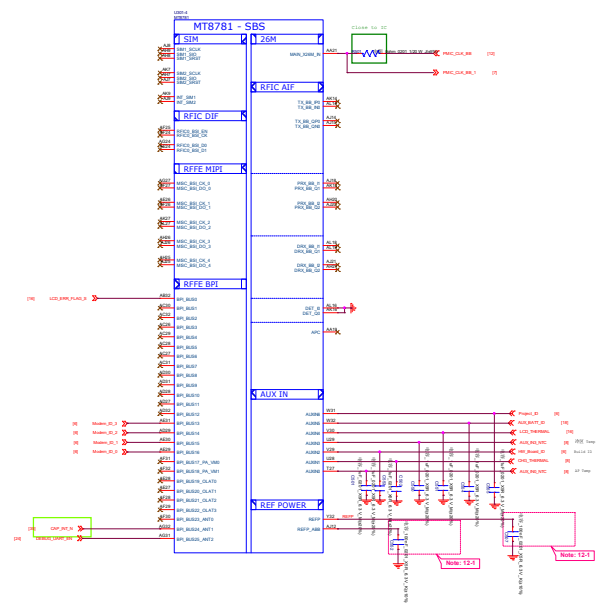
PWRAP_SPD_CSN	AUD_DAT_MOSI0	AP JTAG Mode	IO JTAG Mode
H (Default)	L (Default)	N/A	N/A
H	(by external PU)	N/A	N/A
L	(by external PD)	SPI0_CSB, SPI0_CLK, SPI0_M0, SPI0_M1, EINT13	SPI0_CSB, SPI0_CLK, SPI0_M0, SPI0_M1, EINT14
L	(by external PD)	N/A	N/A
H	(by external PU)	N/A	N/A

Note 12-3: "AUD_SYNC_MOSI" pin features in trapping pin to booting (UFS/leMMC).

AUD_SYNC_MOSI	Storage Booting Mode
L (Default)	Only UFS boot
H	Only eMMC boot
(by external PU)	

Note 12-4: "AUD_DAT_MOSI1" pin features in trapping pin for SW co-load.

AUD_DAT_MOSI1	Mode
L (Default)	LP4x-MCP
H	LP4x-Discrete
(by external PU)	

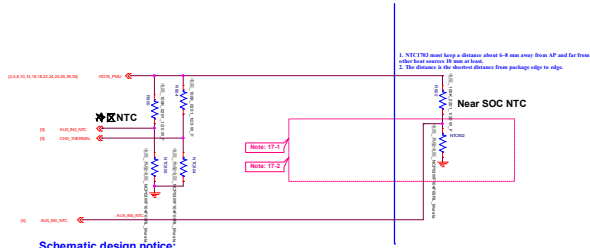


华勤通讯 Huaqin Telecom Technology Com., Ltd

Title 05.BB_1_PMIC_RF

Size D Project L83_MB_V1 Rev v1

Date: Saturday, July 02, 2022 Sheet 5 of 38



Schematic design notice:

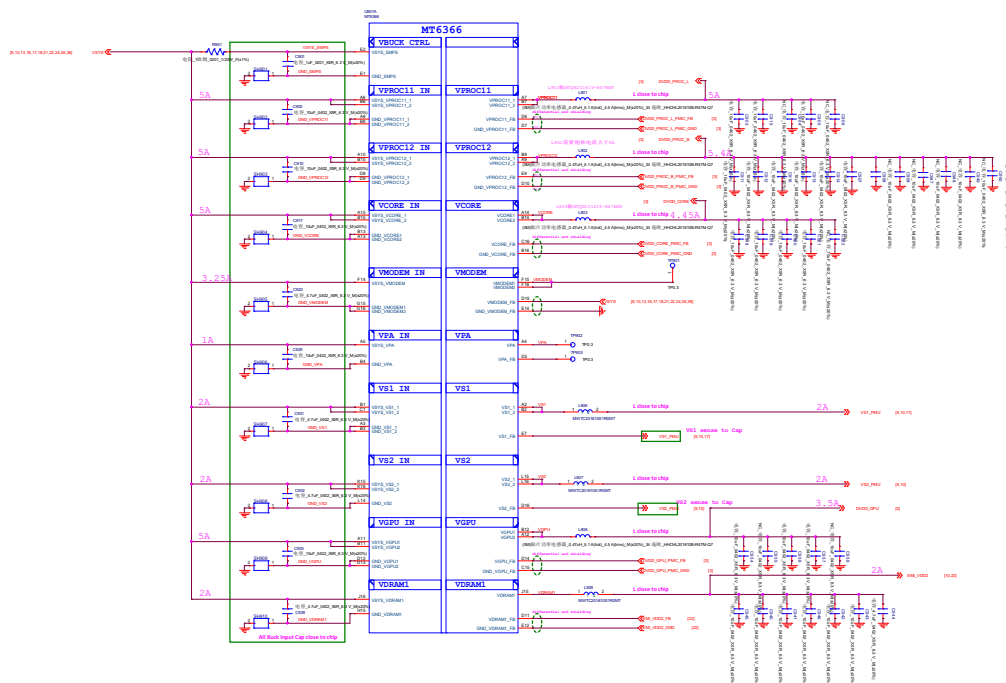
Note 17-1: SW Default Configuration is as followings
AUX_IN2_NTC is for AP Temp.
AUX_IN1_NTC is for MHS PAMMO Temp.

Note 17-2: If your design or placement is different with SW Default Config.
Please refer to "MTXXXX Thermal User Manual.docx" and modify SW setting.

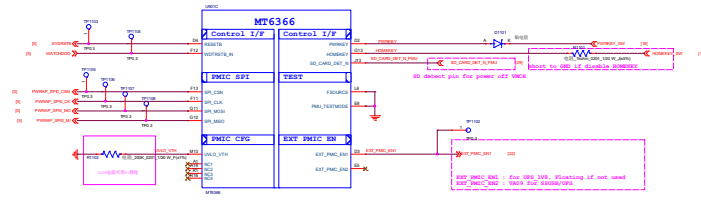
Schematic design notice of "08_BB_AUXADC" page.

	AUX_IN2 (RF SKU/PCB ID) (ADC range is 0.05-1.45V)
SKU1 EU+CHN+JPN	0.05V 0V(R0801+NC, R0802+100K)
SKU2 NA	0.21V 0.21V(R0801+750K, R0802+100K)
Reserved	0.42V 0.41V(R0801+330K, R0802+100K)
Reserved	0.62V 0.62V(R0801+191K, R0802+100K)
Reserved	0.82V 0.82V(R0801+120K, R0802+100K)
Reserved	1.04V 0.99V(R0801+100K, R0802+137K)
Reserved	1.24V 1.24V(R0801+100K, R0802+220K)
Reserved	1.45V 1.45V(R0801+100K, R0802+422K)

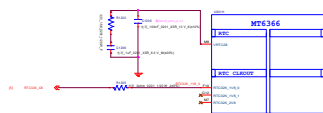
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 08_BB_AUXADC_Thermal		
Size D	Project L83_MB_V1	Rev v1
Date: Saturday, July 02, 2022	Sheet 8 of 38	



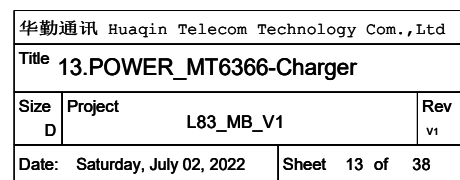
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		09.POWER_MT6366-Buck	
Size	Project	L83_MB_V1	Rev
D			v1
Date:	Saturday, July 02, 2022	Sheet	9 of 38

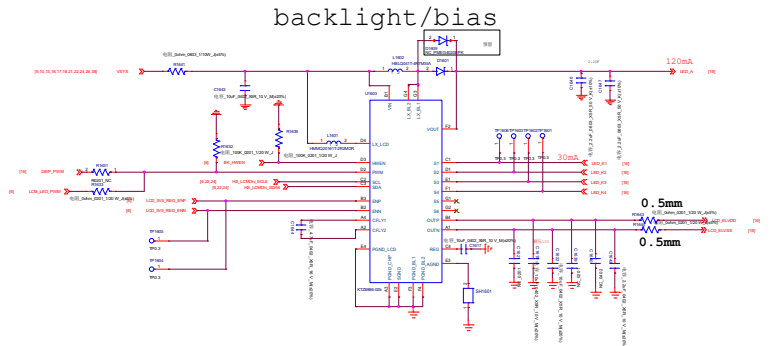
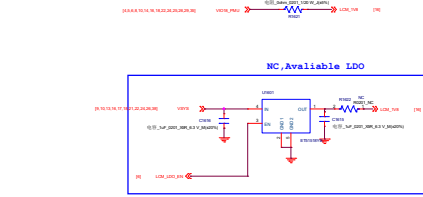
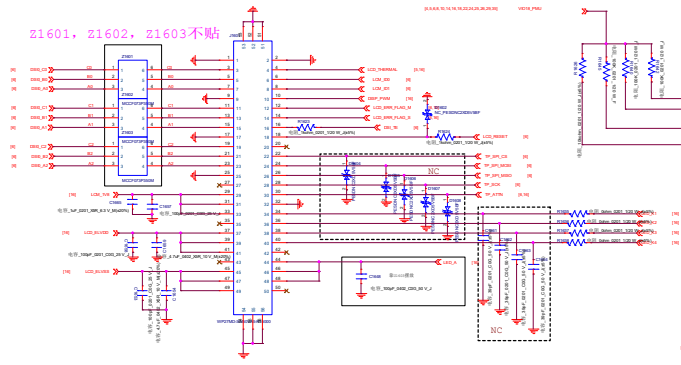


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		11.POWER_MT6366-General	
Size	Project		Rev
D	L83_MB_V1		
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 12.MT6366_SYSCCLK			
Size	Project		Rev
E	L83_MB_V1		v1
Date: Saturday, July 02, 2022		Sheet 12 of 38	



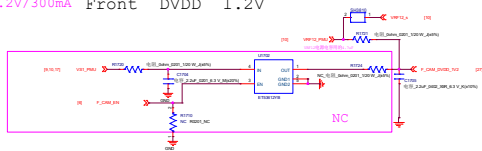


Pin	Signal	IO	Level	Notes
1	VDD	I/O	1.8V	
2	VDDA	I/O	1.8V	
3	VDDIO	I/O	1.8V	
4	VDDIOA	I/O	1.8V	
5	VDD	I/O	1.8V	
6	VDDA	I/O	1.8V	
7	VDDIO	I/O	1.8V	
8	VDDIOA	I/O	1.8V	
9	VDD	I/O	1.8V	
10	VDDA	I/O	1.8V	
11	VDDIO	I/O	1.8V	
12	VDDIOA	I/O	1.8V	
13	VDD	I/O	1.8V	
14	VDDA	I/O	1.8V	
15	VDDIO	I/O	1.8V	
16	VDDIOA	I/O	1.8V	
17	VDD	I/O	1.8V	
18	VDDA	I/O	1.8V	
19	VDDIO	I/O	1.8V	
20	VDDIOA	I/O	1.8V	
21	VDD	I/O	1.8V	
22	VDDA	I/O	1.8V	
23	VDDIO	I/O	1.8V	
24	VDDIOA	I/O	1.8V	
25	VDD	I/O	1.8V	
26	VDDA	I/O	1.8V	
27	VDDIO	I/O	1.8V	
28	VDDIOA	I/O	1.8V	
29	VDD	I/O	1.8V	
30	VDDA	I/O	1.8V	
31	VDDIO	I/O	1.8V	
32	VDDIOA	I/O	1.8V	
33	VDD	I/O	1.8V	
34	VDDA	I/O	1.8V	
35	VDDIO	I/O	1.8V	
36	VDDIOA	I/O	1.8V	
37	VDD	I/O	1.8V	
38	VDDA	I/O	1.8V	
39	VDDIO	I/O	1.8V	
40	VDDIOA	I/O	1.8V	
41	VDD	I/O	1.8V	
42	VDDA	I/O	1.8V	
43	VDDIO	I/O	1.8V	
44	VDDIOA	I/O	1.8V	
45	VDD	I/O	1.8V	
46	VDDA	I/O	1.8V	
47	VDDIO	I/O	1.8V	
48	VDDIOA	I/O	1.8V	
49	VDD	I/O	1.8V	
50	VDDA	I/O	1.8V	
51	VDDIO	I/O	1.8V	
52	VDDIOA	I/O	1.8V	
53	VDD	I/O	1.8V	
54	VDDA	I/O	1.8V	
55	VDDIO	I/O	1.8V	
56	VDDIOA	I/O	1.8V	
57	VDD	I/O	1.8V	
58	VDDA	I/O	1.8V	
59	VDDIO	I/O	1.8V	
60	VDDIOA	I/O	1.8V	
61	VDD	I/O	1.8V	
62	VDDA	I/O	1.8V	
63	VDDIO	I/O	1.8V	
64	VDDIOA	I/O	1.8V	
65	VDD	I/O	1.8V	
66	VDDA	I/O	1.8V	
67	VDDIO	I/O	1.8V	
68	VDDIOA	I/O	1.8V	
69	VDD	I/O	1.8V	
70	VDDA	I/O	1.8V	
71	VDDIO	I/O	1.8V	
72	VDDIOA	I/O	1.8V	
73	VDD	I/O	1.8V	
74	VDDA	I/O	1.8V	
75	VDDIO	I/O	1.8V	
76	VDDIOA	I/O	1.8V	
77	VDD	I/O	1.8V	
78	VDDA	I/O	1.8V	
79	VDDIO	I/O	1.8V	
80	VDDIOA	I/O	1.8V	
81	VDD	I/O	1.8V	
82	VDDA	I/O	1.8V	
83	VDDIO	I/O	1.8V	
84	VDDIOA	I/O	1.8V	
85	VDD	I/O	1.8V	
86	VDDA	I/O	1.8V	
87	VDDIO	I/O	1.8V	
88	VDDIOA	I/O	1.8V	
89	VDD	I/O	1.8V	
90	VDDA	I/O	1.8V	
91	VDDIO	I/O	1.8V	
92	VDDIOA	I/O	1.8V	
93	VDD	I/O	1.8V	
94	VDDA	I/O	1.8V	
95	VDDIO	I/O	1.8V	
96	VDDIOA	I/O	1.8V	
97	VDD	I/O	1.8V	
98	VDDA	I/O	1.8V	
99	VDDIO	I/O	1.8V	
100	VDDIOA	I/O	1.8V	

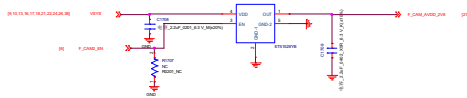


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 16.POWER_MT6371-HV			
Size D	Project	L83_MB_V1	Rev V1
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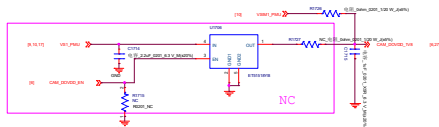
DVDD 1.2V/300mA Front DVDD 1.2V



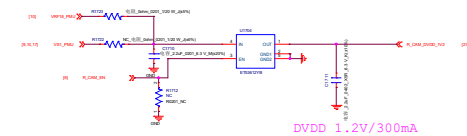
Front AVDD 2.8V 兼容平台供电



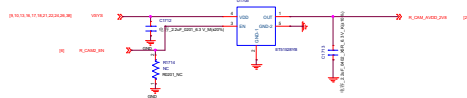
Front DOVDD (Rear DOVDD) 1.8V 兼容平台供电



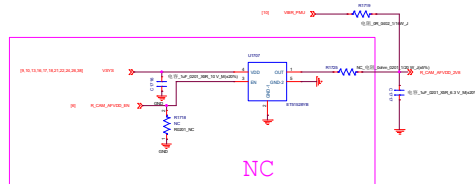
Rear DVDD 1.2V 兼容平台供电



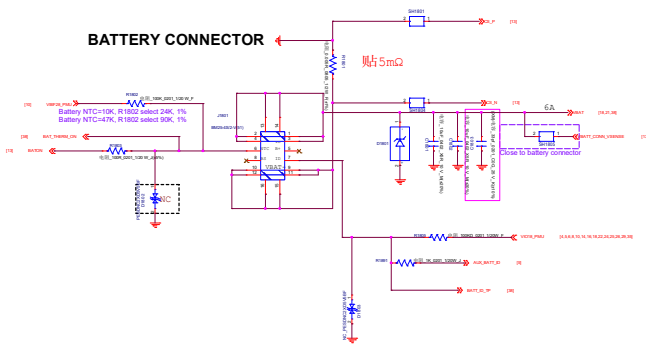
SUB Rear AVDD 2.8V 兼容平台供电



SUB Rear AFVDD 2.8V 兼容平台Vibr供电

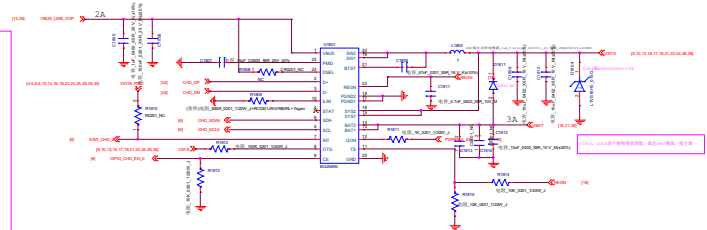


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Title 17.POWER_EXT		
Size D	Project L83_MB_V1	Rev v1
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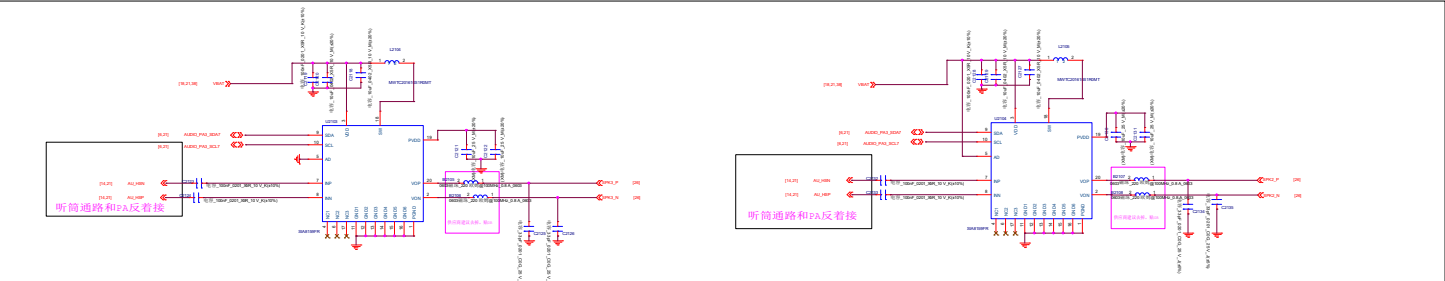
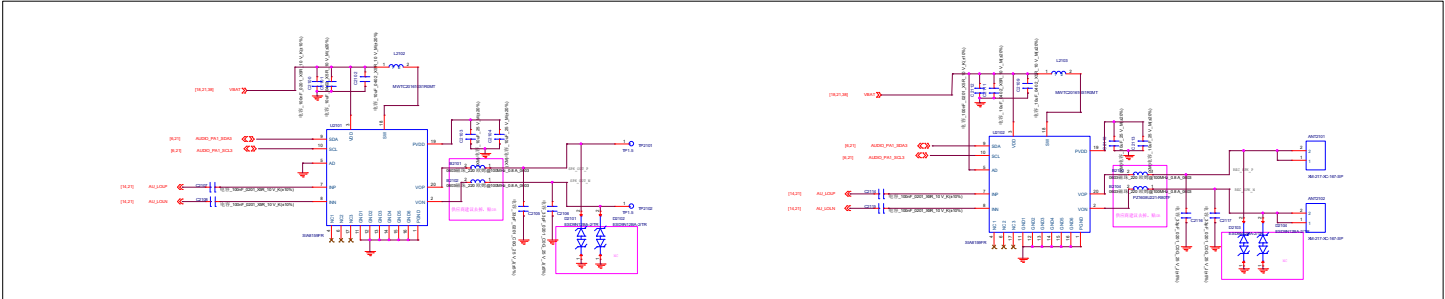


Power Key Spring

SW Charger



华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		18.Charger/CC_LOGIC/BAT_conn	
Size	Project	L83_MB_V1	Rev
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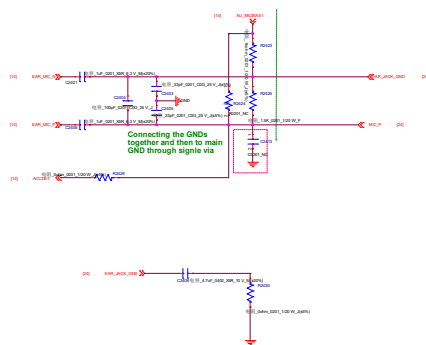


AD pin	A1	A0	PC address (7-bit)
Connects to GND	0	0	0x58
Connects to SCL	0	1	0x59
Connects to SDA	1	0	0x5A
Connects to VDD	1	1	0x5B

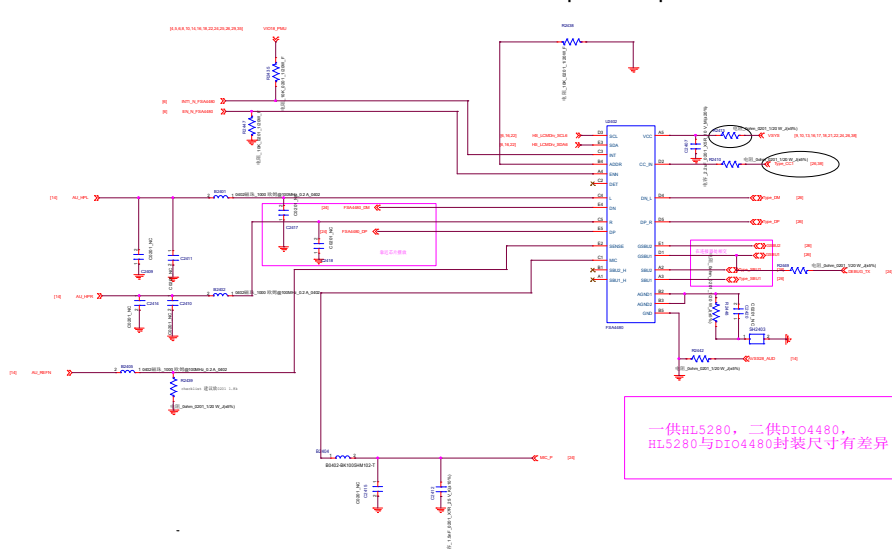
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 21.AUDIO_SPK			
Size D	Project	L83_MB_V1	Rev v1
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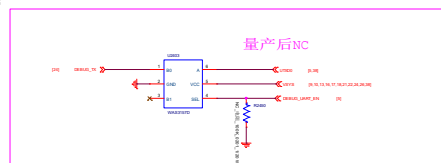
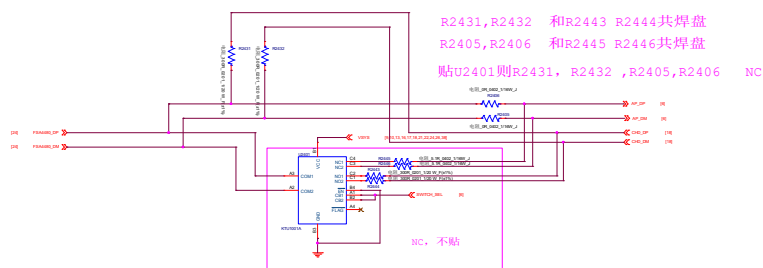
Earphone Mic



Earphone Microphone



DP/DM SW



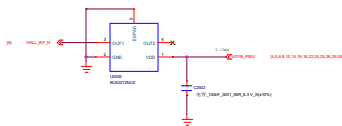
华勤通讯 Huaqin Telecom Technology Com., Ltd

Title 24.MIC_RCV/handset switch

Size D	Project	Rev
	L83_MB_V1	V1

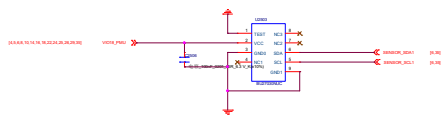
Date: Saturday, July 02, 2022	Sheet 24 of 38
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HALL



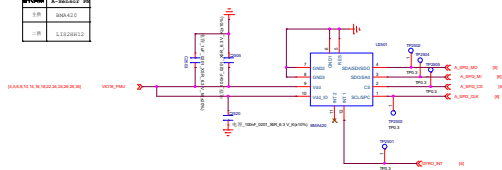
单光感

厂家	IC 型号
ROHM	BH1750

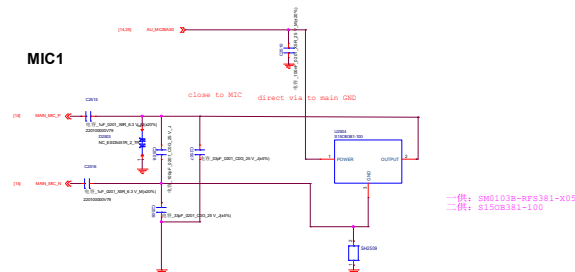


A Sensor

厂家	IC 型号
ROHM	BH1750
IC	BH1750

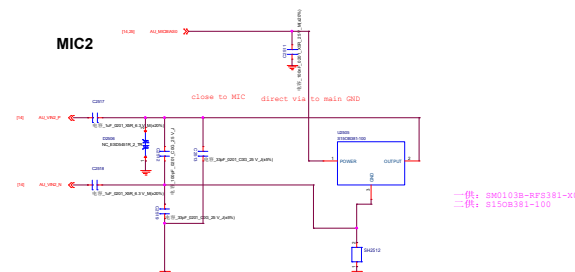


MIC1



一供: SMD103B-RF8381-X05
二供: S150B381-100

MIC2



一供: SMD103B-RF8381-X05
二供: S150B381-100

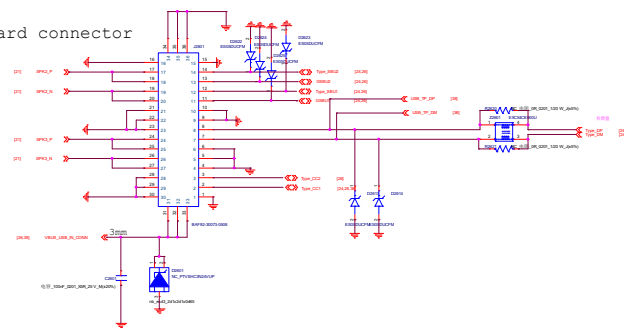
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 25.SENSOR

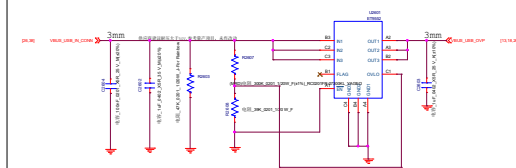
Size D Project L83_MB_V1 Rev V1

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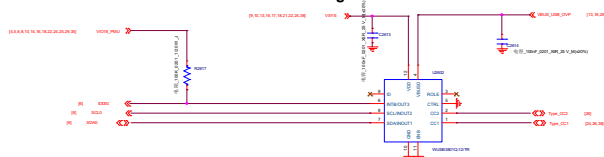
SUB board connector



OVP



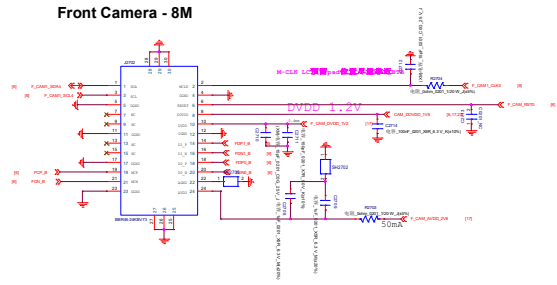
CC Logic



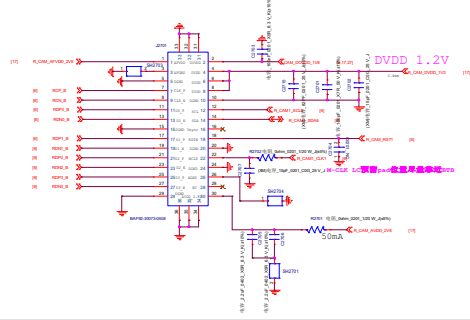
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		26.FP_KEY	
Size	Project	L83_MB_V1	Rev
D			v1
Date:	Saturday, July 02, 2022	Sheet 26 of 38	

★NOTE:

Front Camera - 8M



Main Rear Camera -8M

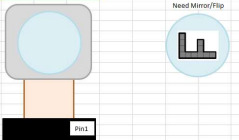


Updated 11/15

1	SDA	2	MCLK
3	SCL	4	D0VDD
5	D0GND	6	RESET
7	NC	8	D0VDD
9	NC	10	D0VDD
11	D0GND	12	D0GND
13	NC	14	D0_P
15	NC	16	D0_N
17	D0GND	18	D0_P
19	MCP	20	D0_N
21	MCP	22	AGND
23	D0GND	24	AVDD

Note:

Standard



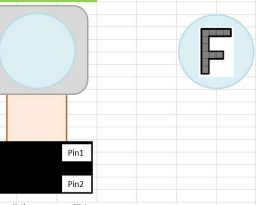
Note: 1. 此为connector图:
2. connector与Lens不同侧
Note: Showing for this picture:
1. Connector means upward.
2. Connector is not faced with Lens field on the same side.

Updated 11/15

1	AVDD	2	D0VDD
3	AFGND	4	D0VDD
5	D0GND	6	D0VDD
7	CLK_P	8	D0VDD
9	CLK_N	10	D0GND
11	D0_P	12	SCL
13	D0_N	14	SDA
15	D0GND	16	Vsync
17	D1_P	18	RSTN
19	D1_N	20	D0GND
21	D2_P	22	MCLK
23	D2_N	24	D0GND
25	D3_P	26	AGND
27	D3_N	28	NC
29	D0GND	30	AVDD_2.8V

Note:

Standard



Note: 1. 此为connector图:
2. connector与Lens不同侧
Note: Showing for this picture:
1. Connector means upward.
2. Connector is faced with Lens field on the same side.

华勤通讯 Huaqin Telecom Technology Com.,Ltd

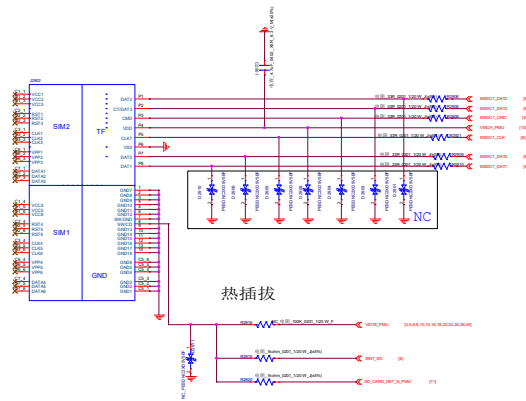
Title

Size Project L83_MB_V1 Rev V1

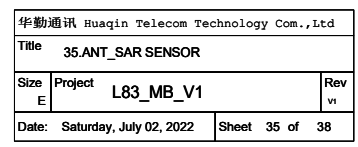
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3IN3 SIM1&SIM2&SD

SD CARD



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		29.SIM_SD_BATT	
Size	Project	L83_MB_V1	Rev
D			V1
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WIFI2.4G/5G_ANT

MT6631

FPC方案

去WIFI测试座

Co-Pad

SWAP方案

接地弹片

POWER

Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note S1-1: For R0015 size, please select 0402 size or larger one.

Note S1-2: Please refer to MT6765 Baseband design notice for VCN33 LDO selection guide.

Note S1-3: If WIFI 5G not support, connect pin 34(WF_RF_5G) to GND.

Note S1-4: Pin 36 (AVDD28_FIM) must be connected to VCN28 even if FM not support.

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Title 36.WCN_MT6631

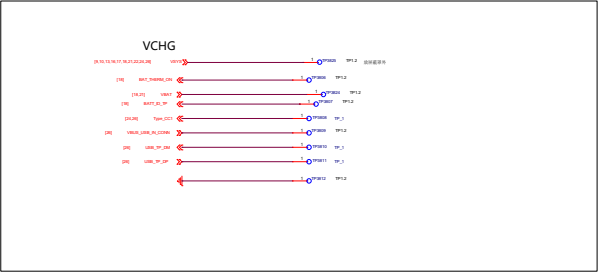
Size D Project L83_MB_V1

Rev v1

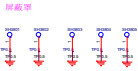
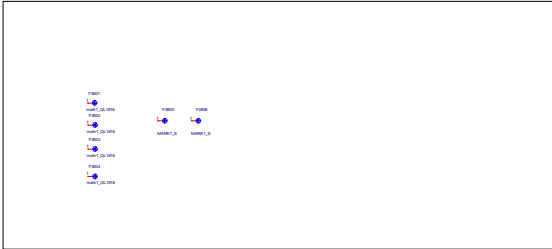
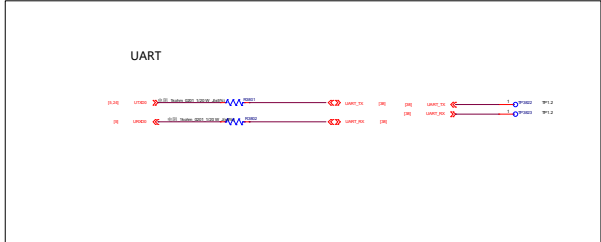
Date: Saturday, July 02, 2022

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华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 37_WCN_ANT		
Size D	Project L83_MB_V1	Rev v1
Date: Saturday, July 02, 2022		Sheet 37 of 38

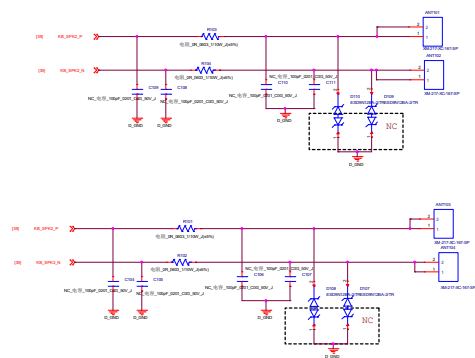


螺丝孔是否接地？

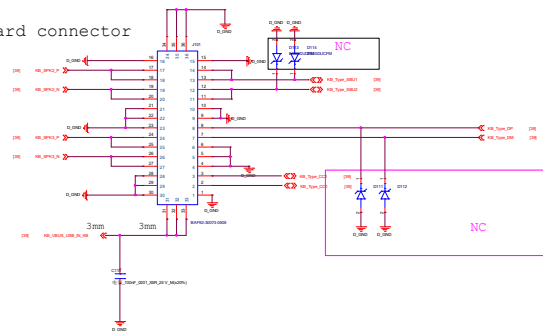


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		38.TEST_POINT_SHIELD	
Size	Project	L83_MB_V1	Rev
D			V1
Date:	Saturday, July 02, 2022	Sheet 38 of 38	

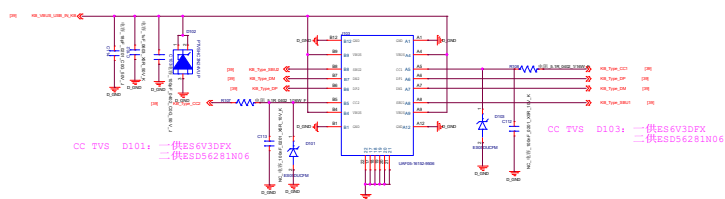
SPK



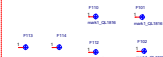
SUB board connector



16 PIN TYPE C Connection



光学定位孔



接地弹片

NC



NAME	NAME
DESCRIPTION	DESCRIPTION
DATE	DATE